

qPCR Data Processing Pipeline

User Guide and Troubleshooting Reference

Scripts covered:

`qpcr_cleaning_pipeline.R` · `qpcr_consolidation.R`

Script version 0.2.0 | For use with CFX Maestro exports

About This Guide

This guide explains how to use the two R scripts that form your laboratory's qPCR data processing pipeline. It is written for biomedical researchers who are comfortable with qPCR experiments and CFX Maestro, but who may not have extensive experience writing or modifying R code.

What the pipeline does: Raw quantitative PCR data exported from CFX Maestro is cleaned, quality-checked, and flagged for manual review according to MIQE-aligned criteria. The cleaning script processes each plate individually and writes cleaned CSV files. The consolidation script then gathers those CSVs into structured Excel workbooks, one sheet per target, ready for downstream analysis.

What you will need to do: Before each run you will need to check a small number of settings at the top of each script, ensure your files are named correctly, and respond to a handful of yes/no prompts in the R console during processing. No programming knowledge is required.

Assumed starting point

R and RStudio are already installed on your computer.

You have run at least one R script before (i.e. you can open a script in RStudio and press Ctrl+Shift+Enter to run it).

Your raw data was exported from CFX Maestro as CSV without any structural modifications to the file (sample names, target names, and Content labels may have been edited, but no columns have been added, removed, or reordered).

Installing Required Packages

Each script requires one or more R packages to be installed before it will run. You only need to do this once per computer. Open RStudio, click on the Console panel at the bottom, and run the following commands by typing them in and pressing Enter:

For both scripts:

```
install.packages(c("tidyverse", "openxlsx"))
```

Installation may take several minutes and will print a lot of text to the console — this is normal. You will see a message like 'package X successfully unpacked' when each package is ready. If you see a red error message saying a package could not be installed, see the Troubleshooting section at the end of this guide.

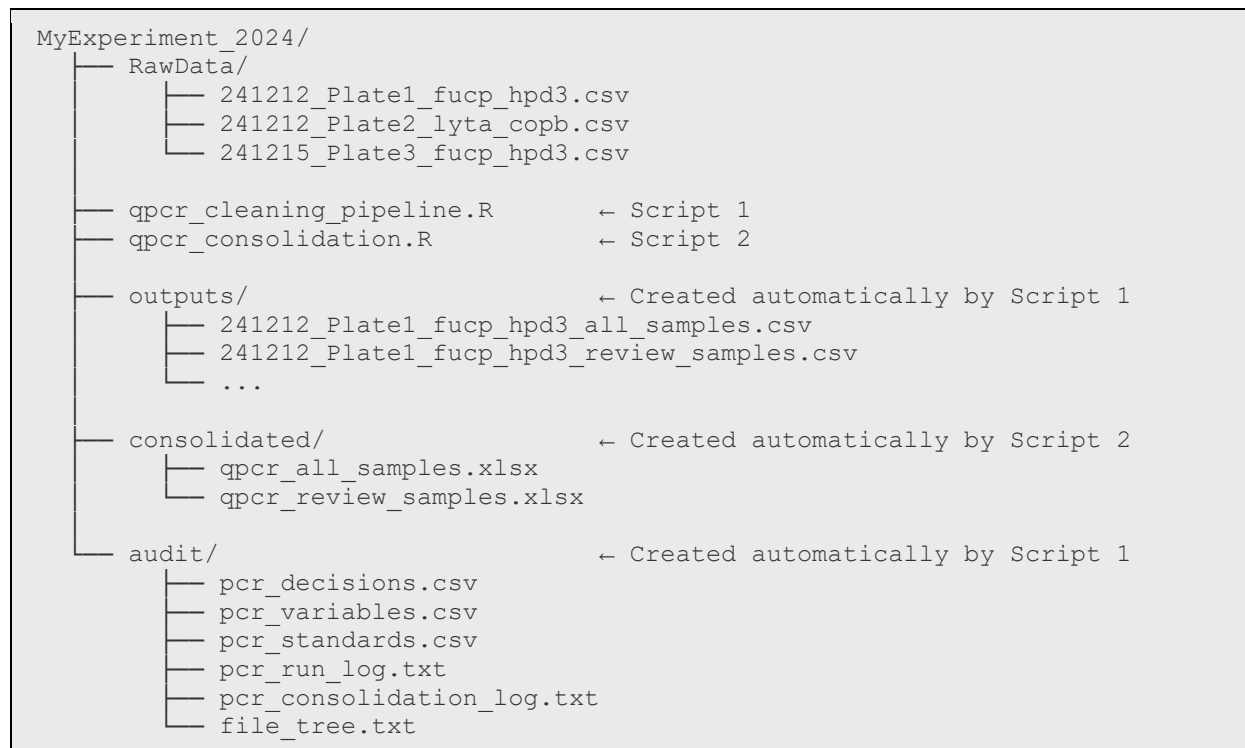
Package version requirements

The script checks that your installed packages are recent enough to run. If you installed R or tidyverse more than two years ago, you may be prompted to update. If an error appears mentioning a minimum version, run the install commands above again — reinstalling updates to the latest version automatically.

Setting Up Your Project Folder

Each experiment should have its own dedicated project folder. The scripts are designed to be placed at the root of this folder, with your raw CFX Maestro CSVs in a subfolder called RawData. When the cleaning script runs, it will automatically create the additional folders it needs (outputs and audit) if they do not already exist.

Recommended folder layout



Before you begin: Open RStudio, then use File → Open Project or File → Open File to navigate to your experiment folder and open the cleaning script (qpcr_cleaning_pipeline.R). Ensure that RStudio's working directory is set to your experiment folder — you should see the correct path in the Files panel on the lower right. If it shows the wrong folder, use Session → Set Working Directory → To Source File Location.

File Naming Conventions

The consolidation script (Script 2) reads the filename of each plate CSV to automatically populate two metadata columns in the Excel output: the run Date and the Plate number. If your filenames do not follow the expected conventions, these columns will be blank in your final workbook — the data itself is unaffected, but you will need to fill in those columns manually after consolidation.

Recommended Filename Format

Name each raw CSV file using the following pattern before placing it in the RawData folder:

```
YYMMDD_PlateN_[target(s)][_optional-info].csv
```

Examples of filenames that will parse correctly:

241212_Plate1_fucp_hpd3.csv	→ Date: 2024-12-12, Plate: 1
20241212_Plate2_lyta_copb.csv	→ Date: 2024-12-12, Plate: 2
241215_Plate3_nuc_rpt2.csv	→ Date: 2024-12-15, Plate: 3, Repeat: 2
241215_plate_no3_lyta.csv	→ Date: 2024-12-15, Plate: 3

Date parsing rules

The date must appear at the very beginning of the filename, followed immediately by an underscore. Both 6-digit (YYMMDD) and 8-digit (YYYYMMDD) formats are accepted.

Format	Example	Parsed as
YYMMDD (recommended)	241212_	2024-12-12
YYYYMMDD	20241212_	2024-12-12
No date at start	Plate1_fucp.csv	(blank — will warn)

Plate number parsing rules

The word 'plate' (case-insensitive) followed by a number should appear somewhere in the filename. Various separators and qualifiers are accepted:

Plate1	Plate_2	Plate-3	plate 4
plate#5	plate no. 6	PlateNo7	

Limitation — not a blocker

If a filename does not contain a parseable date or plate number, a warning will appear in the console during consolidation (e.g. 'No date found in filename: myfile.csv') and those cells will be blank in the Excel output. Your data is still fully processed and consolidated — only those two metadata columns will be empty. You can fill them in manually in Excel after the run, or rename your files to follow the convention before re-running.

Repeat number parsing

If a plate was repeated, you can include 'rpt', 'rep', or 'repeat' followed by a number in the filename. This populates a Repeat# column in the consolidated Excel output:

241215_Plate3_fucp_rpt2.csv	→ Repeat#: 2
241215_Plate3_fucp_rep2.csv	→ Repeat#: 2
241215_Plate3_fucp_repeat2.csv	→ Repeat#: 2
241215_Plate3_fucp_rpt.csv	→ Repeat#: 1 (keyword without number = 1)

Reserved words in sample names — read carefully

The cleaning script automatically removes any row where the Sample column contains the words Std, NTC, or Neg (case-insensitive, checked as a partial match). These are treated as control or standard wells, not experimental samples.

Do not use these words in your real sample names. For example, 'NTC_control', 'Negative_patient', or 'Standard_curve_sample' would all be silently removed from the output. If you have legitimate samples whose names contain these strings, you will need to rename them before running the script.

Examples of names that WOULD be removed:

- Any sample containing 'Std' — e.g. Std-001, Std_High
- Any sample containing 'NTC' — e.g. NTC, NTC_plate2
- Any sample containing 'Neg' — e.g. NegControl, Neg_sample

Data Requirements and Assumptions

This section describes what the script expects to find in your raw CSV files. Understanding these assumptions will help you anticipate any prompts or warnings during processing.

Required CSV Format

Your raw CSV files must be direct exports from CFX Maestro, without any structural modifications. The script dynamically locates the data table inside the file by searching for a row that begins with the word 'Well' — this means CFX Maestro's metadata header rows at the top of the file are handled automatically and do not need to be removed.

Minor edits that are acceptable: Changing sample names, target names, and Content labels within CFX Maestro or directly in the CSV. These are expected and the whole point of the pipeline.

Edits that will break the script: Deleting, renaming, or reordering any of the columns listed below. The script requires these exact column names (it standardises capitalisation and spaces internally, so you do not need to worry about those):

Required column	Description
Well	Well identifier (e.g. A01, B02)
Fluor	Fluorophore channel (e.g. FAM, HEX, ROX)
Target	The gene/target being measured (e.g. fucp, hpd3, uni)
Content	Well type label — includes standard labels (Std-001 to Std-006), NTC, and sample identifiers
Sample	Sample name — your experimental sample identifier
Cq	Quantification cycle value
Starting Quantity (SQ)	Calculated starting quantity from the standard curve

Standards

Before processing any data, the script checks that all six expected standard wells are present in the Content column of each plate. Standards must be labelled as Std-001 through Std-006.

Both label formats are accepted and handled identically:

Label format	Example	Notes
Canonical (preferred)	Std-001, Std-002, Std-003 ...	No warning produced
Abbreviated	Std-01, Std-02, Std-03 ...	Warning in console and audit log, but still processed

Why this matters: Standards Std-001 through Std-003 define the upper limit of detection (LOD Hi), and Std-004 through Std-006 define the lower limit (LOD Lo). If standards are missing, the script cannot reliably apply these thresholds and will ask you how to proceed — this is explained in detail in the interactive prompts section.

Targets and Limits of Detection (LOD)

Each target (gene) in your data has a defined upper and lower limit of quantification. These are set in Section 2 of the cleaning script and are used to assess whether your sample measurements fall within the reliable range of the assay.

LOD boundary	Value	What it means
LOD Hi (upper)	2000 SQ (most targets) 200 SQ (uni/univ/universal)	Maximum reliable SQ from the standard curve. Samples above this are flagged — results are extrapolated and unreliable.
LOD Lo (lower)	0.012 SQ (most targets) 0.0012 SQ (uni/univ/universal)	Minimum detectable SQ. Samples with SQ below this (or no amplification at all) are set to LOD Lo ÷ 2 as a below-detection placeholder.

Critical rule — targets with different LOD values cannot share a plate

The pipeline processes all targets on a plate together and applies a single set of LOD values to all rows. If two targets on the same plate have different LOD Hi or LOD Lo values, the script will stop with an error.

This means: you cannot include both 'uni' (LOD Hi = 200) and 'fucp' (LOD Hi = 2000) on the same plate CSV. These targets must be on separate plate files, or the script will refuse to process the plate.

If you receive an error such as 'Different Hi values found: 200, 2000', check which targets are present in that plate file and separate them before re-running.

Replicates

The script expects each experimental sample to have exactly two replicate wells per target (i.e. duplicates). Samples with one or more than two replicates will still appear in your output, but will be flagged for manual review. See the section on Review Flags for details.

The Universal Target (uni / univ / universal)

uni, univ, and universal are treated as an internal positive control — a target that should amplify in every sample. If any sample completely fails to amplify for this target (no Cq value) or produces a value below the LOD Lo, it is flagged as an 'unexpected negative'. This indicates a potential problem with that sample's DNA quality, quantity, or the presence of inhibitors, and warrants investigation before using any data from that sample.

All three spellings (uni, univ, universal) are recognised as the same target and consolidated onto a single sheet in the Excel output.

Part 1: Running the Cleaning Pipeline

The cleaning pipeline (qpcr_cleaning_pipeline.R) is the first script to run. It reads your raw plate CSV files, applies quality control checks, and writes cleaned output files ready for the consolidation step.

Step 1 — Check your settings (Section 1 of the script)

Open the script in RStudio. Near the top you will find Section 1: Configuration. The settings you may need to change are:

Input directory

```
INPUT_DIR <- "RawData/"
```

This tells the script where to look for your plate CSV files. The default value, `../RawData/`, means 'go up one folder from the script, then into a folder called RawData'. If you are following the recommended project folder structure described earlier, you do not need to change this.

If your CSV files are in a different location, change this path. For example, if they are in a folder called 'Raw' directly inside your project folder, use:

```
INPUT_DIR <- "Raw/"
```

Delta Cq threshold

```
DELTA_CQ_THRESHOLD <- 1.0
```

This controls how large a difference between duplicate Cq values is acceptable. The default of 1.0 means that if your two replicates differ by more than 1 Cq cycle, the pair is flagged for review. This is a commonly used MIQE-aligned threshold. You should not need to change this unless your laboratory uses a different standard.

Skip completed plates

```
SKIP_COMPLETED <- TRUE
```

When set to TRUE (the default), the script checks whether output files already exist for each plate before processing. If both output CSVs already exist, you will be asked whether to skip that plate or reprocess it. This is useful when adding new plates to an existing experiment — you can safely rerun the script without reprocessing plates that have already been cleaned.

Number of standards

```
N_STANDARDS <- 6
```

The number of standard wells expected on each plate (Std-001 through Std-006 by default). Change this only if your standard curve uses a different number of points.

Step 2 — Check your LOD definitions (Section 2 of the script)

Section 2 defines the upper and lower limits of detection for each target. The script comes pre-configured with values for the targets used in your laboratory. You only need to edit this section if you are adding a new target that is not already listed.

If the script stops with an error such as 'Missing targets in LOD_Hi: [targetname]', it means a target in your data is not listed in Section 2. These may be manually added in Section 2. Optionally, you make ask whoever the script maintainers to add it on your behalf

Step 3 — Run the script

With the script open in RStudio, press Ctrl+Shift+Enter (Windows/Linux) or Cmd+Shift+Enter (Mac) to run the entire script. Alternatively, use the Source button in the top-right of the script editor panel.

The script will begin immediately and print progress to the Console panel at the bottom of RStudio. You may be asked to respond to one or more prompts during the run — these are described in detail in the next section.

Console Output Walkthrough

Below is a representative example of what you should expect to see in the R console during a successful run with two plate files. Annotations explain what each section means.

Run header

```
=====
qPCR Pipeline Run Log
-----
Directory   : C:/Users/jsmith/Projects/MyExperiment_2024
Started      : 2024-12-12 09:14:22
User         : jsmith
Input        : RawData/
Output       : outputs
Audit        : audit
=====
```

What this tells you

Confirms the script has started and shows exactly which folder it is reading from and writing to. Check that the Input path points to the folder containing your plate CSVs.

File discovery tree

```

RawData [depth requested: 2 | deepest file: 0 level(s) | 2 file(s) to process]
├─ 241212_Plate1_fucp_hpd3.csv
└─ 241212_Plate2_lyta_copb.csv

File tree saved to: audit/file_tree.txt

```

What this tells you

Lists every CSV file the script found and will process. Check this carefully — if a file you expected is missing, it may be in the wrong folder or have the wrong extension. If an unexpected file appears, it may need to be moved before running.

Standards check (all plates pass)

```

-----
Standards check: all 2 plate(s) passed
Expected: Std-001, Std-002, Std-003, Std-004, Std-005, Std-006
-----

```

All standards present

This is the ideal outcome. All six standard wells were found on every plate and the script proceeds to processing automatically. If the standards check fails for one or more plates, you will see a different prompt — see the next subsection.

Skip completed plates prompt (if SKIP_COMPLETED = TRUE)

```

-----
The following 1 of 2 plate(s) already have both output CSVs:
- 241212_Plate1_fucp_hpd3
-----

Skip these plates and process only the remaining ones?
Enter Y to skip, N to reprocess all: _

```

Y (skip): The already-processed plate is left as-is and only the new plate is processed. Choose this when you are adding new plates to an experiment that was already partially run.

N (reprocess): All plates are processed again from scratch, overwriting any existing output files. Choose this if you have corrected something in a raw data file or changed a script setting.

Per-plate processing

```

[1/2] 241212_Plate1_fucp_hpd3.csv =====
Import      | header at row 19 | 192 data rows
LOD          | Hi=2000          | Lo=0.012      | targets: fucp, hpd3
> Step 0 | Removing unnamed samples ...
Step 0 | done | 0 removed
> Step 1 | SQ adjustment (below LOD_Lo → LOD_Lo / 2) ...
Step 1 | done | 4 adjusted
> Step 2 | Removing matched patterns (Std, NTC, Neg) ...
Step 2 | done | 84 removed (dry_run=FALSE)
> Step 3 | Sample-name consistency check ...

```

```

Step 3 | done | 0 group(s) with mismatched names
> Step 4 | Replicate statistics and review flags ...
Step 4 | done | 2 replicate group(s) flagged for review
> Step 5 | Assembling output tables ...
Step 5 | done

```

```

Output | 108 rows out | 2 for review | 1.3s | 0.012s/row

```

Each step is reported with the number of rows affected. Key things to check:

- **Import** — 'header at row 19' confirms the Well header was found. The number of data rows before any cleaning is shown.
- **LOD** — Confirms which targets were found on this plate and which LOD values are being applied. Verify these match your expectations.
- **Step 1** — Shows how many SQ values were below LOD Lo and set to LOD Lo ÷ 2. A high number here may indicate poor assay sensitivity or inhibited samples.
- **Step 2** — Shows how many rows were removed (standards, NTCs, negative controls). For a standard 96-well plate with duplicates, 6 standards × 2 targets × 2 replicates = 24 standard rows, plus NTCs.
- **Step 4** — Shows how many replicate groups were flagged for review. Zero is ideal but not always achievable; a small number is expected.
- **Output** — Final row count and number of rows flagged for review. The 'rows for review' figure should be small relative to the total.

Run summary

```

=====
Run complete
=====

```

plate	status	rows_in	rows_out	review	elapsed_s
241212_Plate1_fucp_hpd3	processed	192	108	4	1.3
241212_Plate2_lyta_copb	processed	192	110	0	1.1

```

Total rows processed : 218
Total plate time     : 2.4s
Mean rate            : 0.011s/row

```

Run complete

This table gives an at-a-glance check for all plates. If a plate shows 'skipped (failed standards check)' or 'skipped (blank sample names)' in the status column, it was not processed in this run — see the interactive prompts section for how to handle these.

Interactive Decision Points

On some runs the script will pause and ask you a question. These prompts appear because the script has encountered something that requires your judgement — it cannot proceed automatically without your input. This section explains each prompt and what your options mean.

Standards check failure

If one or more plates are missing expected standard wells, the script prints a summary and offers options. The missing standards are classified into three scenarios:

Scenario	What it means	Script label
Both endpoints present, 1–2 interior standards missing	Std-001 and Std-006 are present. Only interior standards (Std-002 to Std-005) are missing. The LOD boundaries can still be inferred.	[MIDDLE ONLY]
Both endpoints present, 3+ interior standards missing	As above but more standards are missing — more uncertainty about the curve shape.	[MIDDLE ONLY — >2 missing]
An endpoint standard is missing	Std-001 (top of curve, defines LOD Hi) or Std-006 (bottom of curve, defines LOD Lo) is absent. The LOD boundary cannot be inferred from the plate.	[ENDPOINT MISSING]

```

-----
Standards check: 1 plate(s) failed (expected Std-001 - Std-006)
Hi half : Std-001 - Std-003 (Std-001 = LOD Hi endpoint)
Lo half : Std-004 - Std-006 (Std-006 = LOD Lo endpoint)
-----
241212_Plate3_fucp [MIDDLE ONLY — 1 interior standard missing, both
endpoints present]
Found   : Std-001, Std-002, Std-004, Std-005, Std-006
Missing : Std-003
Normal LOD : Hi = 2000, Lo = 0.012 (for 'fucp'; from TARGET LOD)
LOD      : LOD Hi affected (missing from Hi half: Std-001-Std-003)

Options:
S = Skip all failing plates (excluded from this run)
M = Allow middle-only plates through with Y/N LOD confirmation;
   enter LOD overrides only for endpoint-missing plates
F = Force all failing plates through (enter LOD overrides for every plate)
Enter S or M or F: _

```

Option S — Skip all failing plates

The plate(s) with missing standards are excluded entirely from this run. They will not appear in the output files. Choose this if you know the plate run was compromised and the data should not be used.

Option M — Mixed handling (recommended for middle-only failures)

Use this when: some plates are missing only interior standards (both Std-001 and Std-006 are present). The script will ask you to confirm or adjust the LOD values for those plates using a Y/N prompt. Plates missing an endpoint standard will require you to manually enter LOD values.

For a plate with middle-only missing standards, you will see:

```

Targets : fucp
LOD Hi   [normal = 2000 | Std-001 present → inferred; applies to all
targets]
→ Is LOD Hi = 2000 correct for this plate? [Y/n]: _

```

Y: Accept the standard LOD Hi value (2000 in the example). Choose this if you have no reason to believe the standard curve for this plate behaved differently from usual.

n: Enter a different value. Use this only if you have inspected the standard curve in CFX Maestro and determined a different upper limit is appropriate.

Option F — Force all plates through

All failing plates are processed, and you are asked to enter LOD values manually for every plate, regardless of which standards are missing. Use this when you have already inspected the standard curves and know what values to use.

Blank sample names

If a plate has no sample names at all in the Sample column (all cells are blank), the script will ask how to proceed:

```
-----
Sample names check: 1 plate(s) have NO sample names
-----
241212_Plate3_fucp
  Rows      : 192
  Content   : Std-001, Std-002, ..., Unknown, NTC
-----
Options:
  S = Skip all    - exclude these plates from this run
  C = Use Content column as Sample name for all these plates
Enter S or C: _
```

Option S — Skip

The plate is excluded from the run. Choose this if the sample names were simply not entered into CFX Maestro and you want to add them before reprocessing.

Option C — Use Content column as Sample name

The Content column (which contains the well-type labels from CFX Maestro, e.g. 'Unknown', 'Std-001') is substituted for the missing Sample name. This allows the plate to be processed, but the resulting sample identifiers in your output will be generic (e.g. 'Unknown') rather than your actual sample IDs. This is logged in the audit trail. Choose this only as a temporary measure — you should update your sample names and reprocess properly.

Part 2: Running the Consolidation Script

After the cleaning pipeline has finished, run the consolidation script (qpcr_consolidation.R). This script reads all the cleaned CSV files from the outputs folder and assembles them into two Excel workbooks — one containing all cleaned data, and one containing only rows flagged for review.

Step 1 — Check your settings (Section 1 of the script)

Input directory

```
INPUT_DIR <- "outputs/"
```

This should match the OUTPUT_DIR from the cleaning pipeline — the default 'outputs/' is correct if you have not changed either script's settings.

Output location

```
CONSOLIDATION_DIR <- "consolidated"
ALL_OUT_PATH <- file.path(CONSOLIDATION_DIR, "qpcr_all_samples.xlsx")
REVIEW_OUT_PATH <- file.path(CONSOLIDATION_DIR, "qpcr_review_samples.xlsx")
```

The two Excel workbooks will be written to a folder called 'consolidated' inside your project folder. You can change the output filenames here if you prefer a different naming convention.

Target alias mapping

```
TARGET_ALIASES <- list(
  uni = c("uni", "universal", "univ")
)
```

This tells the script to treat 'uni', 'universal', and 'univ' as the same target and consolidate their data onto a single Excel sheet named 'uni'. If you have other targets that appear under multiple spellings across different plates, you can add them here in the same format. You do not normally need to change this.

Step 2 — Run the script

With the consolidation script open in RStudio, press Ctrl+Shift+Enter (or Cmd+Shift+Enter on Mac) to run it. The script will print progress to the console as it reads each CSV and builds the Excel workbooks.

The output file conflict prompt

If the output Excel files already exist from a previous run, the script will ask how to handle them:

```
-----  
All-samples workbook already exists:  
Modified : 2024-12-10 14:32:05  
Size      : 243.1 KB  
Sheets    : run_summary, fucp, hpd3, lyta, uni (5 sheets)  
-----
```

```
Options:  
O = Overwrite  - discard existing workbook and rebuild  
A = Append     - merge new data into existing workbook  
N = New file   - save under a different name  
Enter O, A, or N: _
```

Option O — Overwrite

The existing workbook is deleted and rebuilt from scratch using all CSVs currently in the outputs folder. Use this when you have reprocessed all plates or want a completely fresh consolidation.

Option A — Append

New data from freshly processed plates is merged into the existing workbook. Duplicate rows (matched on File Name + Well + Fluor + Target + Sample) are removed automatically so plates that already exist in the workbook are not doubled up. Use this when you are adding new plates to an ongoing experiment.

Option N — New file

The existing workbook is kept untouched and the new consolidation is saved under a different name. You will be prompted to enter a filename suffix, or press Enter to use a timestamp automatically (e.g. qpcr_all_samples_20241215_143200.xlsx).

Consolidation console output

```
-----  
Reading all-samples files  
-----  
Reading [=====] 3/3 (100%) 2.1s elapsed  
[+] all-samples: 326 row(s) loaded from 3 of 3 file(s)  
[~] all-samples: 1 Target value(s) normalised to lowercase: 'FUCP'-'>'fucp'  
  
-----  
Building all-samples workbook - 4 target sheet(s)  
-----  
Sheets [=====] 4/4 (100%) 0.8s elapsed  
[+] Sheet 'fucp' - 124 row(s)  
[+] Sheet 'hpd3' - 88 row(s)  
[+] Sheet 'lyta' - 78 row(s)  
[+] Sheet 'uni' - 36 row(s)  
  
Writing: consolidated/qpcr_all_samples.xlsx  
[+] Saved: qpcr_all_samples.xlsx (287.4 KB)  
  
=====  
Consolidation complete  
=====  
All-samples - 326 row(s) across 4 sheet(s):
```

Sheet	rows
fucp	124
hpd3	88
lyta	78
uni	36

Things to check in the consolidation output

[~] messages (yellow symbol) indicate that something was changed — for example, a target name was lowercased. These are not errors but are worth noting.

[!] messages indicate a warning — for example, a filename did not contain a date or plate number. Review these so you know which metadata columns will be blank.

[x] messages indicate that a file could not be read and was skipped. Investigate why before using the output.

Check that the row counts per sheet are plausible given the number of plates and samples you ran.

Understanding Your Output Files

Cleaning pipeline outputs (in the outputs/ folder)

For each plate CSV the cleaning pipeline processes, two CSV files are written to the outputs folder:

1. [platename]_all_samples.csv

Contains every experimental row from the plate after cleaning — all standards, NTCs, and rows with no sample name have been removed, but all remaining rows are present regardless of whether they passed QC. This is your complete cleaned dataset for that plate.

2. [platename]_review_samples.csv

Contains only the rows that were flagged as needing manual review. This is a subset of the _all_samples file — the same data, filtered to highlight anything that requires your attention before using the results. If a replicate group is flagged, both rows of that pair appear here so you have the full context.

Column descriptions

Both CSV files contain the same columns. The first columns come from your original CFX Maestro export; the last four are added by the cleaning pipeline:

Column	Source	Description
Well	CFX Maestro	Well position (e.g. A01, B02)
Fluor	CFX Maestro	Fluorophore channel (e.g. FAM, HEX)
Target	CFX Maestro	Gene/target name
Content	CFX Maestro	Well type from CFX Maestro (Unkn, Std-001, NTC, etc.)
Sample	CFX Maestro	Your sample identifier
Cq	CFX Maestro	Quantification cycle value. Blank for samples with no amplification, or for groups with more than 2 replicates (see Review Flags).
Starting Quantity (SQ)	Modified by pipeline	SQ value. If the original value was below LOD Lo or absent, this is set to $\text{LOD Lo} \div 2$ as a below-detection placeholder. For groups with more than 2 replicates this is blanked.
DeltaCq	Added by pipeline	The absolute difference between the two replicate Cq values ($ \text{Cq1} - \text{Cq2} $). Only populated on the first row of each replicate pair. Blank if delta Cq cannot be computed (single replicate, excess replicates, or mixed NA/numeric).
AverageSQ	Added by pipeline	Mean of the two replicate SQ values (after any LOD Lo adjustment). Only populated on the first row of each replicate pair.

review_reason	Added by pipeline	A plain-English explanation of why a row was flagged for review. Blank if the row passed all checks. Multiple reasons are separated by semicolons.
---------------	-------------------	--

Review flags — what they mean and what to do

The review_reason column may contain one or more of the following flags. Each flag represents a specific QC concern. For any flagged row, you should inspect the raw data in CFX Maestro before using the result in your analysis.

Flag	What it means	Common causes	Suggested action
DeltaCq exceeds threshold	The two duplicate Cq values differ by more than 1.0 cycle.	Pipetting error, well-to-well variation, partial inhibition in one well.	Check the amplification curves in CFX Maestro. If one curve looks abnormal, exclude that replicate and note the SQ from the other. If both look normal, consider whether the variability is acceptable for your purpose.
Only one replicate present	Only a single well was found for this sample/target combination.	A replicate well was not plated, failed to amplify (see also Mixed Cq flag), or was excluded upstream.	Check the plate layout in CFX Maestro. If the replicate well exists but had no amplification it may appear separately with a Mixed Cq flag. Determine whether the single value is usable or if the sample should be re-run.
One Cq is NA/NaN and the other is numeric	One replicate amplified and one did not.	Partial inhibition, a failed well, insufficient template in one well.	Inspect both wells in CFX Maestro. If the non-amplifying well shows a flat baseline, consider excluding it and using the single Cq value, noting this in your analysis records.
More than 2 replicates — manual reconciliation required	Three or more wells were found for this sample/target combination. Cq and SQ are blanked in the output to prevent accidental use of unchecked values.	A sample was plated in triplicate, plate layout was mislabelled, or two plates with the same sample names were processed together.	REQUIRED ACTION: open the _all_samples.csv and locate the flagged group. Determine which wells are the intended duplicates, remove or reconcile the extra well(s), and enter the correct values manually.

AverageSQ > LOD_Hi	The average SQ for this replicate pair exceeds the top of the standard curve.	Very high bacterial load, or sample was not diluted sufficiently before the assay.	Results above the standard curve are extrapolated and unreliable. Consider repeating the assay with a higher dilution. If the result is only slightly above LOD Hi, note the limitation in your analysis.
Unexpected negative for always-positive target (No amplification)	The universal target (uni/univ/universal) produced no Cq value at all in any replicate.	Degraded DNA, very low DNA concentration, strong inhibitors, or a failed extraction.	This sample's DNA may not be of sufficient quality for reliable quantification of any target. Investigate extraction quality (e.g. NanoDrop or gel) before drawing conclusions from any target for this sample.
Unexpected negative for always-positive target (Below LOD_Lo)	The universal target produced a Cq value but the resulting SQ was below the lower limit of detection.	Very low bacterial DNA, poor extraction, or borderline inhibition.	As above — interpret results for this sample with caution. The DNA is present but at very low levels.
Sample name mismatch within (Fluor, Target, Content)	Two or more different sample names were found for the same Content/Target/Fluor combination. This suggests a labelling inconsistency.	A sample was given different names in different replicate wells, or copy-paste error in CFX Maestro.	Check the sample names in CFX Maestro and correct any inconsistencies. Re-export and reprocess the plate.

Consolidated Excel workbooks (in the consolidated/ folder)

The consolidation script produces two Excel workbooks: qpcr_all_samples.xlsx and qpcr_review_samples.xlsx. Each workbook has the same structure.

Sheet layout

- The first sheet in every workbook is run_summary — a dashboard showing QC counts, standards check outcomes, and sample counts per plate and per target.
- Each subsequent sheet contains data for one target (e.g. fucp, hpd3, lyta, uni). Data from all plates is combined on that sheet.
- Each sheet is formatted as an Excel Table with auto-filters on every column and a frozen header row.

Additional columns in the Excel output

Three columns are added by the consolidation script that are not present in the raw CSVs:

Column	Description
--------	-------------

File Name	The name of the plate CSV file the row came from (e.g. 241212_Plate1_fucp_hpd3_all_samples.csv). Useful for tracing a result back to its original plate.
Date	Run date parsed from the filename (format YYYY-MM-DD). Blank if the filename did not contain a parseable date.
Plate#	Plate number parsed from the filename. Blank if the filename did not contain a plate number.
Repeat#	Repeat number parsed from the filename (e.g. 2 if the filename contained 'rpt2'). Blank if not present.

The run_summary sheet

The run_summary sheet is automatically generated and contains four sections:

1. **Decision Breakdown:** A table showing how many times each QC rule fired for each plate. Columns correspond to rule names (e.g. RV_DELTA_CQ, RV_UNEXPECTED_NEG). High counts for any rule on a specific plate may indicate a systematic problem with that plate.
2. **Standards Check Results:** Shows whether each plate passed the standards check, and if not, what action was taken (passed / skipped / forced through with LOD override).
3. **Final Sample Counts:** A plate × target table showing how many rows are in the all-samples output for each combination.
4. **Review Sample Counts:** The same table but for the review-samples output. Large numbers here relative to the all-samples count warrant attention.

The Audit Folder

Every time the cleaning pipeline or consolidation script runs, it writes to a set of log files in the audit/ folder. These files provide a complete, timestamped record of every decision made during processing. They are intended for verification, troubleshooting, and regulatory compliance purposes — you do not need to interact with them routinely, but understanding what they contain helps you investigate any unexpected results.

pcr_decisions.csv

One row is written for every QC decision made on every sample and target. This is the most detailed audit file and grows with every run.

Column	Description
timestamp	UTC date and time the decision was recorded.
user	The Windows/Mac username of whoever ran the script.
run_id	A unique identifier for the plate processing run (format: YYYYMMDD_HHMMSS_platename). Allows you to filter to a specific run even after multiple re-runs of the same plate.
input_file	The original plate CSV filename.
sample_id	The sample identifier the decision applies to.
target	The target (gene) the decision applies to.
rule_id	The QC rule that was evaluated (e.g. RV_DELTA_CQ, PASS_NEGATIVE). See the Review Flags section for explanations.
outcome	Whether the rule was applied ('applied'), did not fire ('skipped'), or resulted in a clean pass ('pass').
evidence	A brief description of the values that led to the decision (e.g. ' DeltaCq =1.24; thr=1.0').
source	Always 'R_script' — indicates the decision was made automatically.
version	Script version number (0.2.0).

Practical use

To review all decisions for a specific plate, open the file in Excel and filter the input_file column. To see only flagged results, filter outcome to 'applied'. To compare two runs of the same plate, filter by run_id.

pcr_variables.csv

One row is written for each configuration parameter used during a plate's processing run. This creates a permanent record of what settings were in effect — important if thresholds are ever changed between experiments.

Column	Description
timestamp / user / run_id / input_file	Same as pcr_decisions.csv.
var_name	The name of the parameter (e.g. dCq_thr, lod_hi, lod_lo, rm_patterns).
var_value	The value of that parameter at the time of the run.
var_class	The R data type of the value (e.g. numeric, character).

Practical use

If you need to verify what thresholds were applied to a historical plate run, filter by run_id and look at the dCq_thr, lod_hi, and lod_lo rows.

pcr_standards.csv

One row per plate per run, recording the outcome of the standards check.

Column	Description
timestamp / user / run_id / input_file	As above.
n_expected	How many standards were expected (N_STANDARDS setting, default 6).
expected_standards	The full list of expected standard labels (e.g. Std-001 Std-002 ... Std-006), separated by pipes.
found_standards	The standard labels actually found in the plate's Content column.
missing_standards	Standards that were expected but not found. Empty if the plate passed.
action	'pass' — all standards present. 'skipped' — plate was excluded. 'forced' — plate was processed with manually entered LOD values.
lod_override	If action was 'forced', the LOD values that were entered manually. Otherwise blank.
notes	Additional context, for example confirming that abbreviated labelling (Std-01 format) was detected.

pcr_run_log.txt

A plain-text transcript of everything printed to the R console during the cleaning pipeline run. This is overwritten each time the script runs. It captures the full run from file discovery through to the final summary, including any prompts and your responses.

Practical use

If a run produced unexpected results or was interrupted, open this file to review exactly what happened. It is also useful for sharing a run record with a colleague or supervisor without needing them to rerun the script.

If the script exits with an error before completing, the log file may be left in an incomplete state. If subsequent console output stops appearing in RStudio, type `sink()` in the console and press Enter to restore normal output.

pcr_consolidation_log.txt

Equivalent to `pcr_run_log.txt` but for the consolidation script. Records all console output from the consolidation run, including file loading, any warnings about missing dates or plate numbers, and the final workbook write confirmation.

file_tree.txt

A snapshot of the folder structure and file list that was discovered at the start of each cleaning pipeline run. Contains the same ASCII tree that is printed to the console. Useful for confirming exactly which files were included in a given run.

Troubleshooting

This section covers the most common errors and unexpected behaviours. Error messages are shown as they appear in the R console.

Script will not start or package errors

Error: package 'tidyverse' is not installed

```
Error: tidyverse is not installed. Run: install.packages('tidyverse')
```

Run `install.packages("tidyverse")` in the R console and wait for it to complete, then try running the script again.

Error: package version too old

```
Error: dplyr >= 1.1.0 required (installed: 1.0.5).  
Run: install.packages('tidyverse')
```

Your installed version of a package is too old. Run `install.packages("tidyverse")` to update to the latest version.

Console output stops appearing / sink error

Symptom: After the script exits with an error, the R console stops showing output or all output seems to go missing.

Fix: Type the following in the console and press Enter:

```
sink()
```

This closes the log file connection that the script opened. If the problem persists, restart RStudio.

File discovery problems

Error: No matching files found

```
Error: No matching files found.  
INPUT_DIR      : ../RawData/  
FILE_PATTERN   : \.csv$  
Check that the folder exists and contains files matching FILE_PATTERN.
```

The script cannot find any CSV files in the input directory. Check:

- Your working directory in RStudio is set to the project folder (Session → Set Working Directory → To Source File Location).
- The RawData folder exists and contains .csv files.
- The INPUT_DIR setting in Section 1 of the script matches your actual folder name (case-sensitive on Mac and Linux).

Could not find a row with 'Well' in column 1

```
Error: Could not find a row with 'Well' in column 1 of: ../RawData/myplate.csv
Check that the file is a valid CFX Manager export.
```

The script expects CFX Maestro's standard CSV format. This error means the file does not contain the expected 'Well' header row. Possible causes:

- The file was saved from a different software or instrument.
- The file was opened and saved in Excel, which may have corrupted the format — always work with copies, not originals.
- The file is from a different export format within CFX Maestro (ensure you are exporting as 'Quantification Results' CSV).

Missing required columns

```
Error: Missing required columns in: myplate.csv
Expected but not found: starting_quantity_sq
Columns present: well, fluor, target, content, sample, cq
```

A required column is missing or has an unexpected name. The most common causes are:

- The plate was exported without the Starting Quantity column — ensure your CFX Maestro export settings include all columns.
- A column was renamed or deleted after export — always use the original, unmodified export file.

LOD and target errors

Error: Missing targets in LOD_Hi / LOD_Lo

```
Error: Missing targets in LOD_Hi: newgene
```

A target name in your data is not defined in Section 2 of the cleaning script. Ask whoever manages the script to add the new target's LOD values to Section 2, or add it yourself following the existing format.

Error: Different Hi (or Lo) values found

```
Error: Raw data subsetting required. Different Hi values found: 200, 2000
```

This is the 'mixed LOD' error described in the Data Requirements section. Your plate contains targets with different LOD Hi (or LOD Lo) values — for example, 'uni' (Hi = 200) and 'fucp' (Hi = 2000) on the same plate file. The pipeline cannot process these together.

To fix: the plate's CSV data for these targets needs to be split into separate files before running the script. Each file should contain only targets that share identical LOD Hi and LOD Lo values.

Standards check failures

A plate repeatedly fails the standards check

If a plate consistently shows missing standards across multiple runs, check the following:

5. **Content column labelling:** Open the raw CSV and look at the Content column. Standards must be labelled exactly as 'Std-001' through 'Std-006' (or 'Std-01' through 'Std-06' in abbreviated format). Any other labelling — for example 'Standard 1', 'S1', or 'STD001' — will not be recognised.

6. **Number of expected standards:** If your assay uses a different number of standards (e.g. 5 or 8), the N_STANDARDS setting in Section 1 needs to be updated to match.
7. **Standards on a separate plate:** If you run standards on a single plate and apply the curve to multiple plates, those other plates will have no standards. In this case, set STD_CHECK_ENABLED <- FALSE in Section 1 and provide LOD values directly as STD_FORCE_LOD. Consult whoever manages the script for the correct values.

Output file problems

Output file locked / cannot write

```
[write retry 1/5] Could not write to 'Platel_all_samples.csv'. Waiting 3s...
```

The script automatically retries up to 5 times with a 3-second wait between attempts. This usually happens when the output file is open in Excel. Close the file in Excel and the script will write successfully on the next retry. If all 5 attempts fail, close any open Excel files and rerun the script.

Consolidation: 'No matching files found in outputs/'

```
Error: No matching files found in: outputs/  
Check that the cleaning pipeline has been run successfully.
```

The consolidation script cannot find any _all_samples.csv or _review_samples.csv files. Run the cleaning pipeline (Script 1) first, or check that INPUT_DIR in the consolidation script matches OUTPUT_DIR from the cleaning pipeline.

Unexpected results in the output

My sample rows are missing from the output

The most common reasons are:

- **Sample name contained a reserved word:** If the Sample column for those rows contained 'Std', 'NTC', or 'Neg' (case-insensitive), they were removed in Step 2. Check your sample naming.
- **Sample name was blank:** Rows with no sample name are removed in Step 0. Check that all wells in CFX Maestro have sample names assigned.
- **Plate was skipped:** Check the run summary table in the console output — if the plate shows a 'skipped' status, it was excluded due to the standards check or blank sample names.

DeltaCq and AverageSQ are blank for some rows

This is expected behaviour in two situations:

- **Second row of a replicate pair:** DeltaCq and AverageSQ are only populated on the first row of each replicate group to avoid duplication. Look one row up to find the values.
- **More than 2 replicates (RV_EXCESS_REPS flag):** When three or more wells are found for a group, all Cq, SQ, DeltaCq, and AverageSQ values are deliberately blanked. The review_reason column will say 'More than 2 replicates — manual reconciliation required'. You must resolve this manually.

A target I expected is not showing as a sheet in Excel

Check the following:

- The target name in the CSV must match one of the names in Section 2 of the cleaning script. If the name was misspelled in CFX Maestro, it will not match.
- If the target uses an alias (e.g. 'universal' instead of 'uni'), check that TARGET_ALIASES in the consolidation script includes that alias.
- Target names are normalised to lowercase during consolidation. If your CFX Maestro export used mixed case (e.g. 'GyrB' instead of 'gyrb'), the sheet will be named in lowercase — look for it under the lowercase version of the name.

Quick Reference

Standard workflow

8. Export your plate data from CFX Maestro as CSV (Quantification Results format, no structural modifications).
9. Rename each file to follow the convention: YYMMDD_PlateN_[targets].csv and place in the RawData/ folder.
10. Open qpcr_cleaning_pipeline.R in RStudio. Verify that INPUT_DIR points to your RawData folder.
11. Press Ctrl+Shift+Enter to run Script 1. Respond to any interactive prompts in the console.
12. Check the run summary table — confirm all plates show 'processed' status and row counts look reasonable.
13. Open qpcr_consolidation.R. Verify INPUT_DIR is set to 'outputs/'.
14. Press Ctrl+Shift+Enter to run Script 2. Choose O/A/N if prompted about existing workbooks.
15. Open qpcr_all_samples.xlsx and qpcr_review_samples.xlsx from the consolidated/ folder.
16. Review any rows in the review workbook and investigate flagged results before using them in your analysis.

Reserved words — never use in sample names

Reserved word	What it matches (examples)	Effect
Std	Std, Std-001, STD_curve, MyStd	Row removed from output
NTC	NTC, ntc, NTC_plate2	Row removed from output
Neg	Neg, NegControl, Negative_sample, neg	Row removed from output

Review flag summary

review_reason text	Meaning	Urgency
DeltaCq exceeds threshold	Duplicates differ by > 1 Cq	Review
Only one replicate present	Single well only	Review
One Cq is NA/NaN and the other is numeric	Mixed amplification in duplicates	Review
More than 2 replicates — manual reconciliation required	3+ wells for this group; Cq/SQ blanked	Action required
AverageSQ > LOD_Hi	Above standard curve range	Review
Unexpected negative for always-positive target (No amplification)	uni/univ/universal did not amplify	Investigate sample quality
Unexpected negative for always-positive target (Below LOD_Lo)	uni/univ/universal below detection	Investigate sample quality

Sample name mismatch within (Fluor, Target, Content)	Inconsistent sample labelling	Correct in CFX Maestro and reprocess
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Interactive prompts at a glance

Prompt	Options	Choose when
Skip completed plates?	Y = skip, N = reprocess all	Y when adding new plates to an existing experiment. N when you have changed data or settings.
Standards check failed	S = skip plate, M = mixed handling, F = force all	S if plate data is unreliable. M if only interior standards are missing. F if you have manually verified LOD values.
Blank sample names	S = skip plate, C = use Content as sample name	S if you want to add sample names first. C only as a temporary measure.
Output workbook already exists	O = overwrite, A = append, N = new file	O for a fresh run. A when adding new plates to an existing workbook. N to preserve the existing version.